$30,000. Variations in price have to do with whether or not digital circuits are employed, the size of the power supply utilized, and the dynamic range and stability of the circuitry. A typical research grade potentiostat presently costs about $20,000 and thus is accessible as a standard piece of laboratory instrumentation. For certain specific applications, less expensive equipment is available (in the approximately $1,000 to $2,000 range) with top-end machines costing around $50,000.

Electrochemical investigations involve making the sample of interest into one electrode of an electrochemical cell. As such, it is immediately obvious that this technique is limited to materials having excellent to good conductivity. Metallic samples and conducting organic polymers therefore, immediately jump to mind as appropriate samples. What may be less obvious is the application of electrochemical techniques to the characterization of semiconductors; however, this application is historically central to the development of electronic materials and continues to play a key role in semiconductor development to date.

The listing above points to obvious overlaps with other materials-characterization techniques and suggests complimentary strategies that may be of utility. In particular, the characterization of electronic systems using electrical circuit responses (see ELECTRICAL AND ELECTRONIC MEASUREMENT) or electron spectroscopy (see ELECTRON TECHNIQUES) is often combined with electrochemical characterization to provide a complete picture of semiconducting systems. Thus, for example, one might employ solid-state electronics to determine the doping level or (n-type versus p-type character) of a semiconducting sample prior to evaluation of the sample in an electrochemical cell. Likewise, electron spectroscopy might be used to evaluate band-edge energetics or the nature of surface states in collaboration with electrochemical studies that determine interfacial energetics and kinetics. In passing, it is interesting to note that the first "transistor systems" reported by Bell Laboratories were silicon- and germanium-based three-electrode electrochemical cells (Brattain and Garrett, 1955). While these cells never met commercial requirements forcing the development of the solid-state transistor, these studies were critical in the development of our present understanding of semiconductor interfaces.

This chapter considers the most commonly utilized techniques in modern electrochemistry: polarography (i.e., currents induced by a slowly varying linear potential sweep), cyclic voltammetry (i.e., currents induced by a triangular potential waveform), and AC impedance spectroscopy (i.e., the real and imaginary current components generated in response to an AC potential of variable frequency). The characterization of semiconducting materials, the corrosion of metallic interfaces, and the general behavior of redox systems are considered in light of these techniques. In addition, the relatively new use of electrochemistry as a scanning probe microscopy technique for the visualization of chemical processes at conducting and semiconducting surfaces is also discussed.

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Kissinger, P. T. and Heineman, W. R. (eds.). 1996. Laboratory Techniques in Electroanalytical Chemistry. Marcel Dekker, New York.

INTERNET RESOURCES

http://electrochem.cwru.edu/estir/

Web site of the Electrochemical Science and Technology Information Resource (ESTIR), established by Zoltan Nagy at Case Western Reserve. It serves as the unofficial site of the electrochemistry community.

http://www.electrochem.org/

Web site of The Electrochemical Society.

http://www.eggpar.com/index.html

Web site of Princeton Applied Research (a major supplier of electrochemical instrumentation. This site provides a series of "Application Notes."

http://seac.tufts.edu

Web site of the Society of Electroanalytical Chemistry, providing a full compliment of links to electrochemical sites.

ANDREW B. BOCARSLY

CYCLIC VOLTAMMETRY

INTRODUCTION

A system is completely characterized from an electrochemical point of view if its behavior in the three-dimensional (3D) space composed of current, potential, and time is fully specified. In theory, from such phenomenologic data one can determine all of the system's controlling thermodynamic and kinetic parameters, including the mechanism of charge transfer, rate constants, diffusion coefficients, standard redox potentials, electron stoichiometry, and reactant concentrations. A hypothetical $i(E,t)$ (current as a function of potential and time) data set is shown in Figure 1. This particular mathematical surface has been synthesized using the assumption that the process of interest involves a reversible one-electron charge transfer. The term "reversible" is used here in its electrochemical sense, to indicate that charge transfer between the electrode and the redox-active species both is thermodynamically reversible—i.e., the molecule(s) of interest can be both oxidized and reduced at potentials near the standard redox

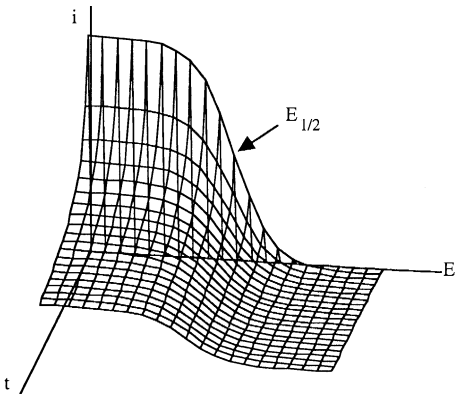


Figure 1. Theoretically constructed current-potential-time [$i(E,t)$] relation for an ideal, reversible, one-electron charge-transfer reaction taking place at an electrode surface. The half-wave potential, $E_{1/2}$, is the redox potential of the system under conditions where reactant and product diffusion coefficents are similar and activities can be ignored.

potential—and occurs at a rate of reaction sufficiently rapid that for all potentials where the reaction occurs, the process is never limited by charge-transfer kinetics. In such a case, the process is completely described by the standard redox potential of the electrochemical couple under investigation, the concentration(s) of the couple components, and the diffusion coefficients of the redox-active species.

On the other hand, one can imagine a redox reaction that is totally controlled by the kinetics of charge transfer between the electrode and the redox couple. In this case, a different 3D current-potential-time surface is obtained that depends on the charge-transfer rate constant, the symmetry of the activation barrier, and the reactant concentrations. Of course, a variety of mechanisms in between these two extremes can also be obtained, each leading to a somewhat different 3D representation. In addition, purely chemical processes can be coupled to charge-transfer events, providing more complex reaction dynamics that will be reflected in the current-potential-time surface.

The enormous information content and complexity of electrochemical dynamics in the current-potential-time domain introduces two major experimental complications associated with electrochemical data. First is the pragmatic consideration of how much time is necessary to obtain a sufficiently complete data set to allow for a useful analysis. The second, far more serious complication is that different charge-transfer mechanisms often translate into subtle changes in the $i(E,t)$ response. Thus, even after one has access to a complete data set, visual inspection is typically an insufficient means to determine the mechanism of charge transfer. Furthermore, knowledge of this mechanism is critical to determining the correct approach to data analysis. Thus, given the requisite data set, one must rely on a series of large calculations to move from the data to chemically meaningful parameters.

The approach described so far is of limited value as a survey tool, or as a global mechanistic probe of electrochemical phenomena. It is, however, a powerful technique for obtaining precise thermodynamic and kinetic parameters once the reaction mechanism is known. The mathematical resolution to the problem posed here is to consider the projection of the $i(E,t)$ surface onto a plane that is not parallel to the i-E-t axis system. This is the basis of cyclic voltammetry. Not surprisingly, a single projection is not sufficient to completely characterize a system; however, if judiciously chosen, a limited set of projections (on the order of a half dozen) will provide sufficient information to determine the reaction mechanism and obtain the key reaction parameters to relatively high precision. Perhaps more importantly, reduction of the data to two dimensions allows one to determine the mechanism based on pattern recognition, and thus no calculational effort is required. Practically, the projection of interest is obtained by imposing a time-dependent triangular waveform on the electrode as given by Equation 1:

$$E(t) = \begin{cases} E_i + \omega t & 0 \text{ for } \leq t \leq t_\lambda \\ E_i + \omega t_\lambda - \omega t & \text{for } t_\lambda \leq t \end{cases} \quad (1)$$

where E_i is the initial potential of the scan, ω is the scan rate (in mV/s), and τ_λ is the switching time of the triangular waveform as shown in Figure 2A. The induced current is then monitored as a function of the electrode potential as shown schematically in Figure 2B. The wave shape of the i-E plot is diagnostic of the mechanism. Different projection planes are simply sampled by varying the value of

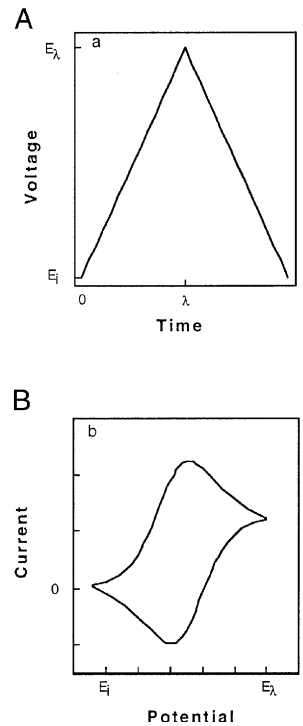


Figure 2. (**A**) Triangular $E(t)$ signal applied to the working electrode during a cyclic voltammetric scan. E_λ is the switching potential where the scan direction is reversed. (**B**) Cyclic voltammetric current versus potential response for an ideal, reversible one-electron redox couple under the $E(t)$ waveform described in (**A**).

ω for a series of cyclic voltammetric scans. Although the 2D traces obtained offer a major simplification over the 3D plots discussed above, their chief power, as identified initially by Nicholson and Shain (Nicholson and Shain, 1964), is that they are quantified from a pattern-recognition point of view by two easily obtained scan-rate-dependent parameters. Nicholson and Shain showed that these diagnostic parameters, peak current(s) and peak-to-peak potential separation, produce typically unique scan-rate dependencies as long as they are viewed over a scan-rate range that traverses at least 3 orders of magnitude. The details of this analysis are provided later. First, one needs to consider the basic experiment and the impact of laboratory variables on the results obtained.

The qualitative aspect of cyclic voltammetry has made it a general tool for the evaluation of electrochemical systems. With regard to materials specifically, the technique has found use as a method for investigating the electrosynthesis of materials (including conducting polymers and redox-produced crystalline inorganics), the mechanism of corrosion processes, the electrocatalytic nature of various metallic systems, and the photoelectrochemical properties of semiconducting junctions. Here a general description of the cyclic voltammetric technique is provided along with some examples focusing on materials science applications.

PRINCIPLES OF THE METHOD

Reaction Reversibility

The Totally Reversible Reaction. The simplest charge-transfer mechanism to consider is given by Equation 2, a reversible redox process:

$$\mathrm{Ox} + n\mathrm{e}^- \Leftrightarrow \mathrm{Red} \tag{2}$$

For this process, the concentration of the reactant and products at the electrode surface is given by the Nernst equation (Equation 3) independent of reaction time or scan rate:

$$E = E_R - (2.303)\frac{RT}{nF}\log\frac{[\mathrm{Red}]}{[\mathrm{Ox}]} \tag{3}$$

where E is the electrode potential, E_R is the standard redox potential of the redox couple, [Red] and [Ox] are the concentrations of the reduced and oxidized species, respectively, R is the gas constant, T is the temperature (K), F is Faraday's constant, and n is defined by Equation 2. [At room temperature, 298 K, 2.303 $(RT/nF) = 59$ mV/n.] Equation 3 indicates that the electrode potential, E, is equal to the redox couple's redox potential, E_R, when the concentrations of the oxidized species and reduced species at the electrode/electrolyte interface are equal. It can be shown that this situation is approximately obtained when the electrode is at a potential that is halfway between the anodic and cathodic peak potentials. This potential is referred to as the cyclic voltammetric half-wave potential, $E_{1/2}$. More precisely it can be demonstrated that the half-wave potential and the standard redox potential are related by Equation 4:

$$E_{1/2} = E_R - \frac{RT}{nF}\ln\left(\frac{\gamma_\mathrm{Ox}}{\gamma_\mathrm{Red}}\left[\frac{D_\mathrm{Red}}{D_\mathrm{Ox}}\right]^{1/2}\right) \tag{4}$$

where D_Ox and D_Red are the diffusion coefficients of the oxidized and reduced species, respectively, and the γ values represent the activity coefficients of the two halves of the couple. While this result is intuitively appealing, a solid mathematical proof of the relationship is complex and beyond the scope of this unit. The mathematical details are well presented in a variety of textbooks, however (Gileadi et al., 1975; Bard and Faulkner, 1980; Gosser, 1993).

By its very nature, the reversible redox reaction cannot cause a substantial change in the connectivity or shape of a molecular system. As a result, the diffusion coefficients of the oxidized and reduced species are expected to be similar, as are the activity coefficents—in which case Equation 4 reduces to $E_{1/2} = E_\mathrm{R}$. Even if there is some variation in the diffusion coefficients, the square root functionality invariably produces a ratio close to 1, and thus the second term in Equation 4 can safely be ignored. Likewise, at a low concentration of electroactive species (as is typically employed in cyclic voltammetry), the activity coefficients can safely be ignored. Once a system has been demonstrated to be reversible (or quasireversible, as discussed later), the redox potential can then be directly read off the cyclic voltammogram.

For the reversible mechanism, the idealized cyclic voltammetric response is illustrated in Figure 2B. Within the context of the Nicholson and Shain diagnostic criteria, this cyclic voltammetric response provides a coupled oxidation and reduction wave separated by $\sim 60/n$ mV independent of the scan rate. The exact theoretical value for the peak-to-potential separation is not critical since this value is based on a Monte Carlo approximation and is somewhat dependent on a variety of factors that are often not controlled. In a real electrochemical cell the ideal value is typically not achieved. The peak height of the anodic and cathodic waves should be equivalent, and the current function (for either the anodic or cathodic waves) is expected to be invariant with scan rate. Under such conditions the peak current is given by Equation 5 (Bard and Faulkner, 1980):

$$i_\mathrm{p} = (2.69 \cdot 10^5)n^{3/2}AD_\mathrm{o}^{1/2}\omega^{1/2}C_\mathrm{o} \tag{5}$$

where n is as defined by Equation 3, A is the area of the working electrode in cm^2, D_o is the diffusion coefficient of the electroactive species having units of cm^2/s, ω is the scan rate in mV/sec, and C_o is the bulk electrolyte concentration in moles/cm^3. (Note that these are not the standard units of concentration.)

One important caveat must be noted here: the peak-to-peak separation is not solely dependent on the charge-transfer mechanism; cell resistances, for example, will increase the peak-to-peak separation beyond the anticipated $60/n$ mV. Thus, the practical application of the diagnostic

is a constant small peak-to-peak separation ($\sim 100/n$ mV) with scan rate. It is also important to realize that the impact of cell resistance on potential ($V = iR$) is scan-rate dependent since the magnitude of the observed current increases with scan rate. Thus, under conditions of high cell resistance (for example, when a nonaqueous electrolyte is used), a reversible couple may yield a scan rate–dependent peak-to-peak potential variation significantly greater than the ideal ~60-mV shift. It is therefore critical to evaluate all three of the diagnostics over a reasonable range of scan rates before reaching a mechanistic conclusion.

The Totally Irreversible Reaction. The opposite extreme of a reversible reaction is the totally irreversible reaction. As illustrated by Equation 6, such a reaction is kinetically sluggish in one direction, producing a charge-transfer–limited current. The reaction is assigned a rate constant k; however, it is important to note that k is dependent on the overpotential, η, where η is the difference between the redox potential of the couple and the potential at which the reaction is being observed ($E_R - E_{electrode}$). It is convenient to define a heterogeneous charge transfer rate constant, k_s, which is the value of k when $\eta = 0$.

$$\mathrm{Ox} \xrightarrow{ne^-} \mathrm{Red} \tag{6}$$

For this case, a mass transport–limited current is only achieved at large overpotentials, since the rate constant for charge transfer is small for reasonable values of the electrode potential. The concentrations of redox species near the electrode are never in Nernstian equilibrium. Thus, one cannot determine either the redox potential or the diffusion coefficients associated with this system. However, careful fitting of the cyclic voltammetric data can provide the heterogeneous charge-transfer rate constant, k_s, and activation-barrier symmetry factor as initially discussed by Nicholson and Shain (1964) and reviewed by Bard and Faulkner (Bard and Faulkner, 1980; Gosser, 1993). In the case of a very small value of k_s, the irreversible cyclic voltammogram is easily identified, since it consists of only one peak independent of the scan rate employed. This peak will shift to higher potential as the scan rate is increased. For moderate but still limiting values of k_s one will observe a large peak-to-peak separation that is scan-rate dependent. The $E_{1/2}$ value will also vary with scan rate. In this case, it is important to rule out unexpected cell resistance as the source of the potential dependence, before concluding that the reaction is irreversible. Often modern digital potentiostats are provided with software that allows the direct measurement (and real time correction) of cell resistance via a current-interrupt scheme in which the circuit is opened for a short (millisecond) time period and the cell voltage is measured. Analog potentiostats often have an iR compensation circuit (Kissinger and Heineman, 1996).

The Quasireversible Reaction. Although arbitrary, it is practically useful to define reversible charge-transfer systems as those having a heterogeneous charge-transfer rate constant in excess of $0.3\omega^{1/2}$, while totally irreversible systems are those exhibiting rate constants less than $2 \times 10^{-5}\omega^{1/2}$. (Note that this definition is quite surprising in that the requisite minimum rate constant for a reversible reaction depends on the potentiostat being employed. As the slew rate of the system increases, the ability to see large charge-transfer rate constants is enhanced. This is unfortunate, in that it clouds the distinction between thermodynamics and kinetics.) This definition produces a large number of reactions having intermediate rate constants ($0.3\omega^{1/2} \geq k_s \geq 2 \times 10^{-5}\omega^{1/2}$), which are referred to as quasireversible systems. These systems will appear reversible or irreversible depending on the scan rate employed. For sufficiently large scan rates, the rate of interfacial charge transfer will be limiting and the system will appear irreversible. For slower scan rates, the system response time will allow the Nernst equation to control the interfacial concentrations and the system will appear reversible. Depending on the scan rate employed one can determine systemic thermodynamic parameters (redox potential, n, and diffusion coefficient) or kinetic parameters. As in the reversible case, the current function for quasireversible systems tends to be scan rate independent. The peak-to-peak potential dependence is also often an inconclusive indicator. However, the potential of the peak current(s) as a function of scan rate is an excellent diagnostic. At low scan rates, the peak potential can approach the theoretical $\sim 30/n$ mV separation from the half-wave potential in a scan rate–independent manner, however, as the scan rate is increased and the system enters the irreversible region and the peak potential shifts with $\log \omega$. This dependence is given by Equation 7 (Gileadi, 1993):

$$E_p^{Irr} = E_{1/2} - \sigma\left[0.52 + \log\left(\frac{D\omega}{\sigma}\right)^{1/2} - \log k_s\right] \tag{7}$$

where σ is the Tafel slope, a measure of the kinetic barrier, and the other terms are as previously defined. Note that the slope of E_p versus $-0.5\log\omega$ allows one to determine (σ/D), while D and $E_{1/2}$ can be obtained from cyclic voltammograms taken in the reversible region allowing one to determine $k(E_{1/2})$ (see Fig. 5C for an example of this type of behavior).

Nonreversible Charge-Transfer Reactions. In contrast to mechanistically "irreversible reactions," which indicate a kinetic barrier to charge transfer, a mechanistically "nonreversible reaction" refers to a complex reaction mechanism in which one or more chemical steps are coupled to the charge-transfer reaction (Nicholson and Shain, 1965; Polcyn and Shain, 1966a,b; Saveant, 1967a,b; Brown and Large, 1971; Andrieux et al., 1980; Bard and Faulkner, 1980; Gosser, 1993; Rieger, 1994). For example, consider the generic chemical reaction shown as Equation 8:

$$\begin{gathered}\mathrm{Ox} + ne^- \Leftrightarrow \mathrm{Red} \\ \mathrm{Red} \xrightarrow{k} \mathrm{Product}\end{gathered} \tag{8}$$

where k represents the rate constant for a nonelectrochemical transformation such as the formation or destruction of

a chemical bond. This reaction couples a follow-up chemical step to a reversible charge-transfer process and is thus referred to as an EC process (electrochemical step followed by a chemical step). Consider the effect of this coupling on the cyclic voltammetric response of the Ox/Red system. At very fast scan rates, Ox can be converted to Red and Red back to Ox before any appreciable Product is formed. Under these conditions, the Nicholson and Shain diagnostics will appear reversible. However, as the scan rate is slowed down, the redox couple will spend a longer time period in the Red state and thus the formation of Product will diminish the amount of Red below the "reversible" level. Therefore, at slow scan rates the ratio of peak currents (i_a/i_c) will increase above unity (the reversible value). On the other hand, the current function for the cathodic current will decrease below the reversible level as Red is consumed and thus becomes unavailable for back-conversion to Ox. The peak-to-peak potential is not expected to change for the mechanistic case presented. This set of diagnostics is unique and can be used to demonstrate the presence of an EC mechanism. Likewise, one can consider mechanisms where a preliminary chemical step is coupled to a follow-up electrochemical step (a CE mechanism) or where multiple chemical and electrochemical steps are coupled. In many of the cases that have been considered to date, the Nicholson and Shain diagnostics provide for a unique identification of the mechanism if viewed over a sufficiently large scan-rate window. An excellent selection of mechanisms and their expected Nicholson and Shain diagnostics are presented in Brown and Large (1971).

PRACTICAL ASPECTS OF THE METHOD

Since a cyclic voltammetric experiment is fundamentally a kinetic experiment, the presence of a well-defined internal "clock" is essential. That is, the time dependence is provided experimentally by the selected scan rate since the input parameter $E(t)$ is implicitly a time parameter. However, in order for this implicit time dependence to be of utility it must be calibrated with a chemically relevant parameter. The parameter used for this purpose is the diffusion of an electroactive species in the electrolyte. Thus, the cyclic voltammetric experiment is fundamentally a "quiet" experiment in which convective components must be eliminated and the diffusion condition well defined. This is done by considering the geometry of the electrochemical cell, the shape of the electrode under investigation, and the time scale of the experiment.

Additionally, from the essence of the experiment, the application of an electrode potential combined with the monitoring of the induced current, it is important to know to high precision the time-dependent electrode potential function, $E(t)$, for all values of t. This capability is established by using a potentiostat to control an electrochemical cell having a three-electrode cell configuration. Successful interpretation of the electrochemical experiment can only be achieved if the experimenter has a good knowledge of the characteristics and limitations of the potentiostat and cell configuration employed.

Electrochemical Cells

Cyclic voltammetric experiments employ a "three-electrode" cell containing a working electrode, counterelectrode (or auxiliary electrode), and reference electrode. The working electrode is the electrode of interest; this electrode has a well-defined potential for all values of $E(t)$. The counterelectrode is simply the second electrode that is requisite for a complete circuit. The reference electrode is in fact not an electrode at all, but rather an electrochemical half cell used to establish a well-defined potential against which the working electrode potential can be measured. Typically, a saturated calomel electrode (SCE—a mercury-mercurous couple) or a silver–silver chloride electrode is utilized for this purpose. Both of these reference half cells are commercially available from standard chemical supply houses. While the consideration of reference half cells is important to any electrochemical experiment, it is beyond the scope of this unit; the reader is referred to the literature for a consideration of this topic (Gileadi, 1993; Gosser, 1993; Rieger, 1994). The exact geometry of the three electrodes, along with the shape and size of the working electrode, will determine the internal cell resistance and capacitance. These electrical parameters will provide an upper limit for both the current flow (via Ohm's law: $i = V/R$, where V is voltage and R is resistance) and the cell response time (via the capacitive time constant of the cell). The contact between the reference half cell and the remainder of the electrochemical cell is typically supplied through a high-impedance frit, ceramic junction, or capillary. A high-impedance junction is employed for two purposes: to eliminate chemical contamination between the contents of the reference electrode electrolyte and the electrolyte under investigation and to guarantee a minimal current flow between the reference and working electrodes; the voltage drop associated with this current, referred to as iR drop (Ohm's law), is uncompensated by standard potentiostat circuitry. If this voltage drop becomes substantial (see Equation 9 and discussion below), the data will be highly distorted and not representative of the chemical thermodynamics/kinetics under investigation.

Potentiostats and Three-Electrode Electrochemical Cells

At its heart, the potentiostat is a negative-feedback device that monitors the potential drop between the working electrode and the reference electrode. If this value deviates from a preselected value, a bias is applied between the working and counterelectrodes. The working/counterelectrode voltage drop is increased until the measured working electrode versus reference electrode potential returns to the preset value (Gileadi et al., 1975; Gileadi, 1993). In order to carry out a cyclic voltammetric experiment, a potential waveform generator must be added to the potentiostat. The waveform generator produces the potential stimuli presented by Equation 1. Depending on the instrument being utilized, the waveform generator may be internal to the instrumentation or provided as a separate unit.

The cyclic voltammetric "figure of merit" of a potentiostat will depend on the size of the power supply utilized

and the rise time (or slew rate) of the power supply. The rise time determines the maximum scan rates that can be used in fashioning the $E(t)$ waveform. In addition to the power-supply slew rate, the cell requirements of a nonconvective system coupled with practical limits at which the potential of the working electrode can be varied (due to the resistance-capacitance (RC) time constant associated with a metal/electrolyte interface) provide upper and lower limits for the accessible scan-rate range. Unless the cell is carefully insulated from the environment, thermal and mechanical convective currents will set in for scan rates much below ~1 to 2 mV/s; this establishes a practical lower limit for the scan rate. The upper limit is dependent on the size of the electrode and the resistance of the cell. For typical cyclic voltammetric systems (electrode size ~1 mm^2), scan rates above ~10 V/s tend to introduce complications associated with the cell RC time constant. However, scan rates as high as 100,000 V/s have been reported in specially engineered cells employing ultramicroelectrodes and small junction potentials. More realistically, careful limitation of cell resistance allows one to achieve maximum scan rates in the 10 to 100-V/s range.

The size of the power supply determines the potentiostat compliance voltage (the largest voltage that can be applied between the working and counterelectrodes). This voltage determines the potentiostat's ability to control the potential of the working electrode. If one is working in a single-compartment cell with an aqueous electrolyte containing a relatively high salt concentration (≥0.1 molar salt), then a relatively modest power supply (~10 V) will provide access to all reasonable working electrode potentials. However, in order to carry out cyclic voltammetric studies in high-resistance cells (i.e., those with low salt concentrations, multiple electrochemical compartments, and/or nonaqueous electrolytes) a compliance voltage on the order of 100 V may be required. It is extremely important to note when using a high-compliance voltage potentiostat that the potential reported by the potentiostat is the voltage drop between the reference electrode and the working electrode. Although this value will never exceed ~3 V, in order to achieve this condition, the potentiostat may have to apply ~100 V between the counter and working electrodes. *Since the leads to these electrodes are typically exposed, the researcher must use utmost care to avoid touching the leads, even when the potentiostat is reporting a low working electrode potential. If the experimenter is accidentally inserted into the circuit between the working and counterelectrodes, the full output voltage of the power supply may run through the experimenter's body.*

In order to determine the potential of the working electrode with regard to the reference electrode, the potentiostat employs an electrometer. The value measured by the electrometer is used both to control the potentiostat feedback loop and to produce the potential axis in the cyclic voltammogram. As such it is assumed that the electrometer reading accurately reflects the potential of the working electrode, ϕ. In fact, the electrometer reading, E_{obs}, is better represented by Equation 9:

$$E_{obs} = \phi + iR \tag{9}$$

where iR is the uncompensated voltage drop between the working and reference electrodes. In general, application of a voltage between two electrodes (in this case, the working and reference) may cause a current to flow. This would result in a large value for the iR term (and be deleterious to the reference half cell). This problem is circumvented by employing a high-impedance junction between the working electrode and reference electrode, as noted earlier. This junction ensures that the value of i will be quite small and thus iR will have a small value. In this case $E_{obs} \approx \phi$ and the potentiostat reading is reliable. However, if R is allowed to become excessively large, then even for a small value of i, the iR term cannot be ignored, and an error is introduced into the cyclic voltammogram. High scan rates produce large peak currents, not only exacerbating the iR drop but introducing a phase-lag problem associated with the cell's RC time constant. Both these effects can severly distort the cyclic voltammogram. If this occurs, it can be remedied by switching to a low-impedance reference electrode. Decreasing the cell R improves both the iR and RC responses at the cost of destabilizing the potential of the reference half cell.

The Working Electrode

The working electrode may be composed of any conducting material. It must be recognized that the shape, area, and internal resistance of this electrode affect the resulting current (Nicholson and Shain, 1964; Bard and Faulkner, 1980; Kissinger and Heineman, 1996), and therefore these parameters must be controlled if the cyclic voltammetric data is to be used analytically. Almost all of the kinetic modeling that has been carried out for cyclic voltammetric conditions assumes a semi-infinite linear-diffusion paradigm. In order to meet this condition, one needs a relatively small electrode (~1 mm^2) that is planar. The small size assures low currents. This both limits complications associated with iR drop and provides well-defined diffusion behavior. If the above conditions cannot be met, then a correction factor can be employed prior to data analysis (Kissinger and Heineman, 1996).

In cases where cyclic voltammetric studies are applied to solution species, typical working electrode materials include platinum, gold, mercury, and various carbon materials ranging from carbon pastes and cloths to glassy carbon and pyrolytic graphite. These materials are selected because they are inert with respect to corrosion processes under typical electrochemical conditions. A second selection criteria is the electrode material's electrocatalytic nature (or lack thereof). Platinum, broadly speaking, presents a catalytic interface. This is particularly true of reactions involving the hydrogen couple. As a result, this material is most often utilized as the counterelectrode, thereby ensuring that this electrode does not kinetically limit the observed current. For the same reasons, platinum tends to be the primary choice for the working electrode. The other materials noted are employed as working electrodes because they either are electrocatalytic for a specific redox couple of interest, provide exceptional corrosion inertness in a specific electrolyte of interest, or present a high overpotential with respect to interfering redox couples. With

respect to this latter attribute, carbon and mercury are of interest, since both have a high overpotential for proton reduction. As such, one can access potentials significantly negative of the water redox potential in aqueous electrolyte using a carbon or mercury working electrode. Typically, potentials that are either more negative than the electrolyte reduction potential or more positive than the electrolyte oxidation potential are not accessible, since the large current associated with the electrolysis of the electrolyte masks any other currents. Carbon also presents an interesting interface for oxidation processes in aqueous electrolytes since it also has a high overpotential for water oxidation. Mercury, on the other hand, is not useful at positive potentials since it is not inert in this region, oxidizing to Hg^{2+}. In addition to standard metal-based electrodes, a variety of cyclic voltammetric studies have been reported for conducting polymer electrodes, semiconducting electrodes, and high-T_c superconducting electrodes.

An alternate basis for selecting an electrode material is mechanical properties. For example, the mercury electrode is a liquid electrode obtained by causing mercury to flow through a glass capillary. The active electrode area is at the outlet of the capillary, where a droplet of mercury forms, expands, and eventually drops off. This has two major electrochemical consequences. First, the electrode surface is renewed periodically (which can be helpful if surface poisoning by solutions species is an issue); second, the area of the electrode increases in a periodic manner. One can control (as a function of time) the electrode area and lifetime by controlling the pressure applied to the capillary. Obviously, this type of control is not available when using a solid electrode.

Similarly, certain solid materials offer unique mechanical characteristics providing enhanced electrode properties. For example, carbon electrodes are available in a variety of crystalline and bulk morphological forms. Unlike many materials carbon is available as a fiber, woven cloth, or paper. The availability of carbon as a paper has recently become important in the search for new electrode materials using combinatorial materials techniques. Large-area carbon-paper electrodes can be used to hold a well-defined library of potential new electrode materials made by constructing a stoichiometric distribution of alloys of two or more materials as described by Mallouk et al. (Reddington et al., 1998) Each alloy electrode is prepared as a small dot on a carbon-paper electrode. Each carbon-paper electrode holds a grid of >100 dots that provide a gradient of stoichiometries over the chemical phase-space being investigated. A survey marker is then needed to determine which dots are catalytic when the whole carbon sheet is used as an electrode. (In Mallouk's case a fluorescent pH indicator was used to identify electrodes that experienced a large pH shift in the electrolyte at modest potentials.) Potential new electrocatalyst formulations can then be cut out of the carbon grid and mounted as individual electrodes, for which a complete electroanalysis can then be carried out. In this manner a wide expanse of compositional phase-space can be searched for new electrocatalytic materials in a minimum amount of time. As a result, complex alloys containing three or more constituents can be evaluated as potential new electrode materials.

Electrolytes

The basic requirements of an electrolyte are the abilities to support the flow of current and to solubilize the redox couple of interest. In the context of a cyclic voltammetric experiment solubility becomes central, both because of the resistance limitation expressed by Equation 9 and because of the need to maintain a diffusion-limited current. It is most useful to consider the electrolyte as composed of three components, a solvent system, a supporting electrolyte, and an electroactive species. The electroactive species is the redox couple of interest. In the case of a species in solution, the species is typically present in millimolar concentrations. Below ~0.1 mM, the current associated with this species is below the sensitivity of the cyclic voltammetric experiment. At concentrations above ~10 mM it is difficult to maintain a diffusion-limited system.

The low concentration of electroactive material employed in the cyclic voltammetric experiment precludes the presence of sufficient charge carriers to support the anticipated current. Therefore, a salt must be added to the system to lower the electrolyte resistivity. This salt, which is typically present in the 0.1 to 1 M concentration range, is referred to as the supporting electrolyte. The supporting electrolyte cannot be electroactive over the potential range under investigation. Since this substance is present in significantly higher concentrations than the electroactive species, any redox activity associated with the supporting electrolyte will mask the chemistry of interest.

Water, because of its high dielectric constant and solubilizing properties, is the electrolyte solvent of choice. However, organic electrolytes are often needed to solubilize specific electroactive materials or provide a chemically nonreactive environment for the redox couple under investigation. Because of its good dielectric and solvent properties, acetonitrile is a primary organic electrolyte for cyclic voltammetry. Acetonitrile will dissociatively dissolve a variety of salts, most typically perchlorates and hexafluorophosphate-based systems. Tetraalkylammonium salts are often the supporting electrolytes of choice in this solvent, with tetra-*n*-butylammonium perchlorate—which dissolves well in acetonitrile, is electrochemically inert over a large potential window, and is commercially available at high purity—being a popular option. One important limitation of the acetonitrile systems is their affinity for water. Acetonitrile is extremely hydroscopic, and thus water will be present in the electrolyte unless the solvent is initially dried by reflux in the presence of a drying agent and then stored under an inert gas. Extremely dry acetonitrile can be prepared using a procedure published by Billon (1959). A variety of other solvents have been employed for specific electrochemical applications. The types of solvents available, compatible supporting electrolytes, solvent drying procedures, and solvent electrochemistry are nicely summarized by Mann (1969).

DATA ANALYSIS AND INITIAL INTERPRETATION

As noted earlier, one of the major attributes of cyclic voltammetric experiments is the ability to determine the

charge-transfer mechanism with relatively little data analysis using a pattern-recognition approach. In accord with the procedures established by Nicholson and Shain, initial data analysis is achieved by considering the response of current and peak-to-peak potential shifts as a function of the potential scan rate (Nicholson and Shain, 1964; Bard and Faulkner, 1980; Gosser, 1993). Three diagnostic plots were suggested by these authors:

A plot of current function, $(i_p/\omega^{1/2})$;
A plot of peak-to-peak potential separation versus scan rate;
A plot of the ratio of the anodic peak current to cathodic peak current versus scan rate.

These three diagnostic plots have become the cornerstones of cyclic voltammetric analysis. The current function required for the first plot is the peak current divided by the square root of the scan rate. This functional form is related to the fact that in the absence of kinetic limitations, the rate of charge transfer is diffusion controlled. Fick's second law of diffusion introduces a $\omega^{1/2}$ (where ω is the scan rate) dependence into the current response. Thus, division of the peak current by $\omega^{1/2}$ removes the diffusional dependence from the cyclic voltammogram's current response.

Both the first and third diagnostic require measurement of peak currents. This represents a special challenge in cyclic voltammetry due to the unusual nature of the baseline. This is most easily seen by the representation in Figure 3, which shows an idealized cyclic voltammogram plotted both in its normal *i-E* form and as i versus t. Recall that both forms are mathematically equivalent since E and t are linearly related by Equation 1. Note that the first wave observed starting at a potential of E_i has a peak current that is easily obtained (line 1), since the baseline is well established. However, it is important to realize that the peak current for the return wave is not given by line 2. This is easily seen by looking at the time-dependent plot. Here it can be seen that the return wave has a baseline that is very different from the initial wave, and thus the measurement of peak current is complex. In addition, it has been noted that the baseline established for the return wave is a function of the switching potentials. Several approaches have been suggested to resolve this baseline problem. Nicholson (1966) has suggested a mathematical solution that produces the true peak current based on the values of lines 1, 2, and 3 in Figure 3A. Using this approach the ratio of peak currents is obtained as Equation 10:

$$\frac{i_c}{i_a} = \frac{\text{line 1}}{\text{line 2}} + 0.485\frac{\text{line 3}}{\text{line 4}} + 0.086 \qquad (10)$$

where i_c and i_a are the cathodic and anodic peak currents respectively.

A critical, and often-forgotten, aspect of the Nicholson and Shain analysis is that it is necessary to observe the diagnostics over 3 orders of magnitude in scan rate in order to come to a definitive conclusion. This is often difficult to do. While in many cases one can still reach reasonable mechanistic conclusions, the uncertainty in the conclusions is increased as the scan rate range is limited. Certainly, determination of a charge transfer mechanism using less than 2 orders of magnitude in scan rate cannot be justified. Since a system which is chemically reactive may change with time, it is useful to randomize the order of scan rates employed. This way, the time dependence of the system will not be confused with the scan rate dependence.

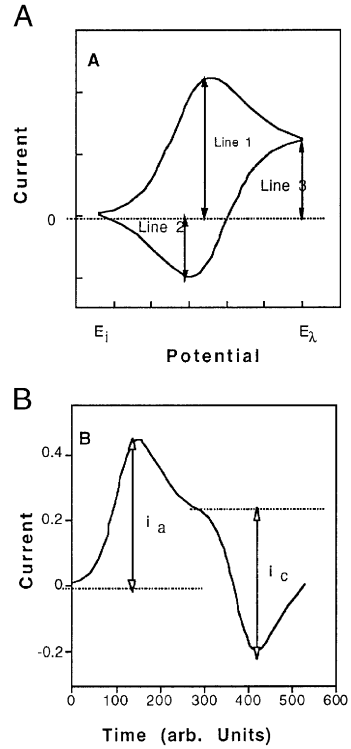


Figure 3. (**A**) Ideal one-electron reversible cyclic voltammogram showing measurements from an $i = 0$ baseline. (**B**) Same as (**A**) but showing the time dependence of the data and identifying the true baselines for the anodic and cathodic cyclic voltammetric peaks.

An Example

One mechanism that has received much attention in recent years is the electrocatalytic mechanism. In its simplest form it can be expressed as in Equation 11:

$$\begin{aligned} &\text{Ox} + n\text{e}^- \Leftrightarrow \text{Red} \\ &\text{Red} + Z \xrightarrow{k\Leftrightarrow} \text{Ox} + \text{Product} \end{aligned} \qquad (11)$$

The electrocatalytic mechanism involves an initial charge transfer from the electrode to Ox followed by a bimolecular solution charge transfer between Red and a dissolved reactant (Z) having a rate constant k. This follow up step regenerates Ox, so that the Ox/Red couple is never consumed. This mechanism is often referred to as

"mediated charge transfer." It allows one to circumvent an interfacial activation barrier for the direct oxidation of Z at the electrode. In addition, it tends to endow the reaction with a degree of specificity, since the interaction of Z with Red can be tailored by one's choice of molecular system. As such, this mechanism forms the basis for many modern electrochemical sensors. One example of this application is given in Figure 4, which demonstrates a sensor for the detection of the amino acid sarcosine as a clinical marker for kidney function. Sarcosine is unreactive at solid metal electrodes. The experiment presented here utilizes a graphite electrode and a phosphate buffered (pH = 7.0) electrolyte containing 6×10^{-6} molar sarcosine oxidase (SOX), a naturally occurring enzyme that selectively oxidizes sarcosine according to the reactions shown in Equation 12. The redox mediator in this case is $[Mo(CN)_8]^{4-}$ that has been bonded to the electrode surface. The sensor mechanism is a variation on the mechanism shown in Equation 11. Note that in this case, the oxidized form of the mediator is the active species.

$$\begin{aligned} [Mo(CN)_8]^{4-} &\Leftrightarrow [Mo(CN)_8]^{3-} + e^- \\ [Mo(CN)_8]^{3-} + SOX &\rightarrow [Mo(CN)_8]^{4-} + SOX^+ \\ SOX^+ + \text{Sarcosine} &\rightarrow SOX + \text{Sarcosine}^+ \\ \text{Sarcosine}^+ &\rightarrow \text{chemical products} \end{aligned} \qquad (12)$$

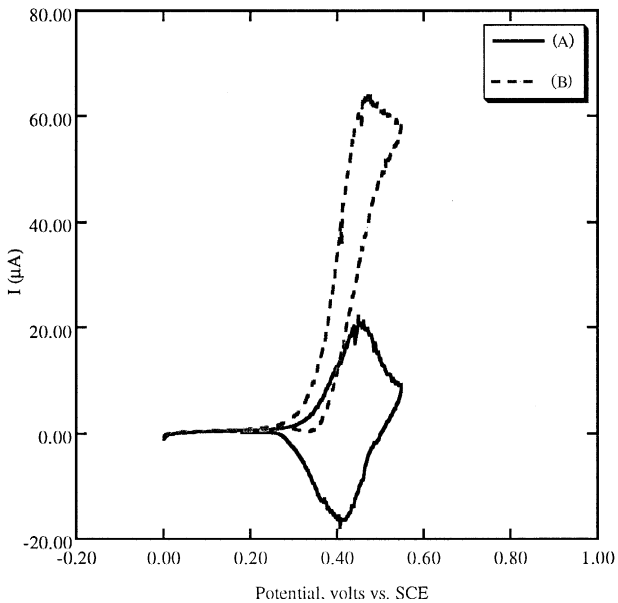


Figure 4. Cyclic voltammograms of a graphite electrode coated with a $Mo(CN)_8^{4-}$-containing polymer in an aqueous electrolyte containing 0.05 M phosphate buffer (pH = 7), 0.35 M KCl supporting electrolyte, and 6 μM sarcosine oxidase. A scan rate of 2 mV/s was employed, with the scan being initiated at 0.00 V versus SCE and proceeding to positive potentials. Scan (**A**) shows a quasireversible wave for the one-electron oxidation of the Mo complex. Scan (**B**) is the same as (**A**) with the addition of 50 mM sarcosine. The increase in the anodic wave and decrease in the cathodic wave are indicative of a mediated charge-transfer mechanism. This electrode system has been used to analyze for creatinine, a species involved in kidney function.

Scan (a) in Figure 4 shows the cyclic voltammetric response of the $[Mo(CN)_8]^{4-}$ system. Because of its low concentration, the addition of sarcosine oxidase has no perceptible effect on the cyclic voltammetric response. The electrocatalytic chain shown in Equation 12 is activated by the addition of the analyte, sarcosine, as shown in scan (b). The response is highly selective for the presence of sarcosine due to the selectivity of both the $[Mo(CN)_8]^{4-}$ mediator and the enzyme. In addition the system is very sensitive to low concentrations of sarcosine. The shift from a reversible system, scan (a), to an electrocatalytic system, scan (b), is trivially monitored by cyclic voltammetry, nicely illustrating the qualitative utility of this technique in determining reaction mechanisms. Note that both the disappearance of the cathodic wave and the enhancement in the anodic wave are requisite for the mediation mechanism (Equation 11).

Finally, it should be noted that the exact scan-rate dependence observed will be a function of k, the chemical rate constant for all coupled processes. Hence, one can obtain this quantity from a quantitative analysis of the scan-rate dependence. Thus, for chemically coupled electrochemical systems, cyclic voltammetry can be utilized as a kinetic tool for determination of the nonelectrochemical rate constant. Classically, this analysis has involved curve fitting data sets to the numerically derived functional forms of the cyclic voltammetric wave shape. This approach has been well documented and discussed from both theoretical and applied viewpoints by Gosser (1993). A more powerful kinetic analysis is available by comparison of data sets to digitally simulated cyclic voltammograms. Until recently this approach required extensive programming skills. However, digital simulation programs are now commercially available that allow one to import a cyclic voltammetric data set and correlate it with a simulated voltammogram based on a proposed mechanism (Feldberg, 1969; Speiser, 1996). Both Princeton Applied Research (PAR) EG&G and BioAnalytical Systems (BAS) offer digital simulators that interface with their data-collection software or operate on a freestanding PC data file. Such programs run on personal computers and bring high-powered kinetic and mechanistic cyclic voltammetric analysis to all users. The power of cyclic voltammetry, however, still resides in the fact that mechanistic and kinetic/thermodynamic data for a wide variety of systems can be directly obtained with little effort from raw data sets via the pattern-matching processes highlighted in this review.

Another Example

The cyclic voltammetric response of a real system is shown in Figure 5. The inorganic compound under study is shown in Figure 5A. The molecular system consists of two iron sites sandwiching a central Pt(IV). Each of the iron sites is composed of a ferrocyanide unit, $Fe(CN)_6^{4-}$, a well-studied, one-electron charge-transfer couple. The compound is held together by two bridging cyanide (CN^-) ligands. The study presented here involved an aqueous electrolyte composed of 0.1 M $NaNO_3$ and 2.5 mM electroactive complex. A platinum working electrode and an SCE reference

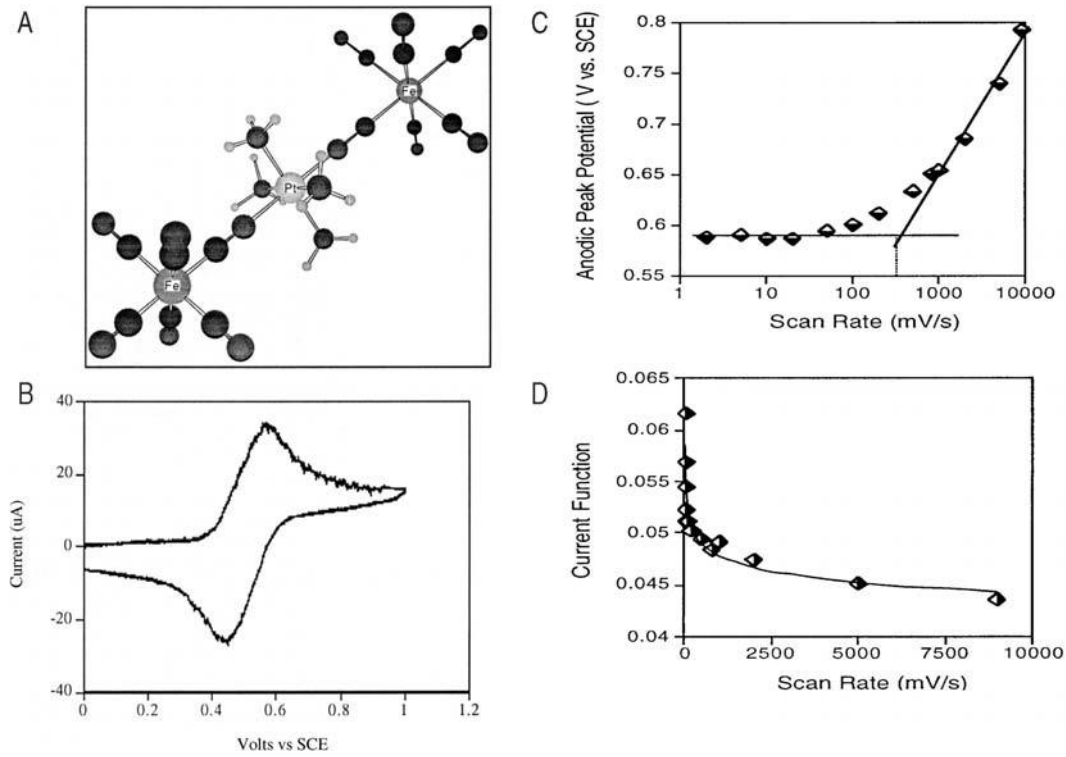


Figure 5. (**A**) Model of $\{[Fe(II)(CN)_6]-[Pt(IV)(NH_3)_4]-[Fe(II)(CN)_6]\}^{4-}$ (**B**) A cyclic voltammogram of the complex in part (**A**) taken at 100 mV/s using a platinum working electrode. The scan initiates at 0.0V versus SCE and proceeds in the positive direction. (**C**) Shift in anodic peak potential as scan rate is increased, showing reversible behavior <300 mV/s (no scan rate dependence) and overall quasireversible behavior. From this data a heterogeneous charge-transfer rate constant at $E_{1/2}$ of 4.6×10^{-3} cm/s is determined. (**D**) Current function for the data set showing the pattern for a catalytic charge-transfer process (see Equation 11).

half cell were employed. Initially, a cyclic voltammetric scan at a rate of 100 mV/s was obtained (shown in Fig. 5B). The presence of a symmetric set of waves yielding identical peak currents (once the baseline correction was applied) yielded the preliminary conclusion that this may be a reversible or quasireversible system. The symmetric nature of the cyclic voltammogram allowed determination of a half wave value, $E_{1/2} = 0.51$ V versus SCE and assignment of this value as the standard redox potential for the couple under investigation. The presence of only one pair of cyclic voltammetric waves along with the observed redox potential indicated that the signal is associated with one or more iron centers, and therefore that the Pt(IV) is electrochemically silent in this system. The value of n was determined by adding an equimolar concentration of hexameineruthenium(II), an ideally reversible one-electron couple, to the cell. The peak current for the Fe/Pt complex was 2.8 times that observed for the hexamineruthenium(II) complex (Zhou et al., 1990). Based on Equation 7, a value of $n = 2$ was deduced and the cyclic voltammetric wave was assigned to the simultaneous redox reaction of both irons. Note that in theory one could directly derive n from either the peak-to-peak potential separation or the peak current (Equation 7). However, the peak-to-peak potential separation at 100 mV/s is 120 mV, suggesting the system is not purely reversible. While Equation 7 might still be employed, a knowledge of the diffusion coefficent is required. Thus, the use of an internal standard represents the best method for obtaining n. Note, the standard must be chosen so that $D^{1/2}$ of the standard and the unknown are comparable.

To further develop the analysis, a plot of E_p for the anodic wave versus logω (Fig. 5C) was undertaken, clearly indicating a quasireversible system with reversible behavior exhibited below a scan rate of 300 mV/s. Construction of the current-function diagnostic indicates a further complexity in the reaction mechanism (Fig. 5D). This behavior is predicted for an electrocatalytic mechanism. In the present case, it has been shown that the observed mechanism is associated with the oxidation of tetraaminoplatinum(II) cations by the oxidized complex to regenerate the reduced complex and form a coordination polymer. The peak-to-peak potential diagnostic similarly supports this conclusion.

Based on this analysis one predicts that a coordination polymer can be synthesized as a thin layer on an electrode by holding the electrode potential at a value positive with respect to the anodic peak potential in the presence of an electrolyte containing a high concentration of the Fe/Pt complex and $[Pt(NH_3)_4]^{2+}$. This conclusion has been

experimentally verified (Pfennig and Bocarsly, 1991). The product obtained is photoactive and can thus be patterned (Wu et al., 1995). It has potential applications as an electrochemical sensor material for alkali cations (Bocarsly et al., 1988; Coon et al., 1998).

SAMPLE PREPARATION

The working electrode can be composed of any material that has a satisfactory conductivity. Thus, electrodes have been fabricated from metals, inorganic semiconductors (both small and large band gap), and conducting organic polymers. Materials in these classes have been electrochemically examined in a variety of bulk forms, including single crystals, polycrystalline solids, or particulates in a paste or pressed pellet. Conducting liquids such as mercury and gallium are also good working electrode candidates. In certain instances nonconducting materials can also be examined electrochemically using the chemically modified electrode approach. For example, the coordination polymer mentioned previously (see Another Example under Data Analysis) can be formed as an ultrathin layer (<1 μm thick) on an electrode surface using electrosynthesis techniques. The redox properties of this insulating solid can be studied under these conditions. Likewise, various nonconducting crystalline materials having redox activity can be studied by first electrocrystallizing microcrystallites of the material onto the surface of an appropriate conducting substrate.

In some cases, it is advantageous to acquire the optical properties of a material while it is undergoing oxidation or reduction. In this case, reflectivity studies can be undertaken by utilizing an optically polished working electrode. Alternatively, the material can be deposited as a thin film on an optically transparent conductor such as tin oxide, ITO (indium tin oxide) or a fine metal mesh. In this case, the optical transmission properties of the material can be evaluated. This latter approach also allows for spectroelectrochemical studies of nonconducting redox thin films. These approaches have found utility in the characterization of semiconductors and conducting organic polymers.

Independently of the material being studied, the size and shape of the working electrode must be taken into account. The cyclic voltammetric response can be distorted if too much current is passed, since this will introduce an unacceptably large iR drop (see Equation 9). Limitation of the electrode area is one of the fundamental methods for limiting current without introducing new resistances. In addition, the capacitance of the electrode-electrolyte interface is proportional to electrode area. This capacitance introduces a phase shift in the current-potential response, which distorts the cyclic voltammogram. Thus, it is critical to minimize electrode area. Pragmatically, the area A should be in the range of 10^{-3} cm$^2 \leq A < 0.1$ cm^2). In cases where the electrode material is fairly resistive (i.e., semiconductors), areas of ~ 1 cm^2 can be successfully employed. In all cases, it is very important to have a counterelectrode that has a substantially larger area than the working electrode. If this condition is not maintained, it is possible for the counterelectrode to become the current limiting electrode, thereby invalidating the cyclic voltammetric diagnostic results.

The mathematically modeled response of the electrode to a triangular potential waveform, which forms the basis for cyclic voltammetric mechanistic studies, has been developed for a spherical electrode. Further, because of the system's dependence on diffusion through the electrolyte, the current-potential response is dependent on the shape of the electrode. Fortunately, the dependence is not strong, and thus can often be ignored. Electrodes that are discs or cylinders can be employed with little negative effect. Often, square planar electrodes are also acceptable; however, in this case, care must be taken that edge effects are not perturbing the current signal. If the electrode is mounted in an insulating material so that a portion of the sample is exposed to electrolyte, it is important that the electrode/insulator interface be totally impervious to electrolyte. If a small amount of electrolyte is held between the electrode and the mount, an enhanced current will be achieved in this area, which will dramatically perturb the electrochemical response. Finally, it should be noted that electrode shape does become critical as the electrode size decreases, thus for electrodes in the 1 to 100-μm length regime, shape and size become major parameters in determining the dynamic response of the system.

SPECIMEN MODIFICATION

In some cases, the cyclic voltammetric properties of a system are dependent on the mechanical conditions of the surface. This is most evident when analyzing single-crystal semiconducting materials, which tend to show crystal-plane dependent charge-transfer properties. This effect has also been reported for single-crystal metal electrodes. Specimen surface modification can be accomplished using standard metallurgical techniques such as abrasive (or chemical) polishing, chemical etching, or electrochemical etching. Since surface damage (either inadvertent or deliberate) can modify the electrochemical response, the condition of the working electrode surface must be monitored. Standard chemical etches are often employed to remove damage during sample preparation and electrode mounting. Chemical etches can also be used to remove native oxides from a sample's surface. Such oxides are often semiconducting or insulating; thus, they modify the sample's electrochemical response.

PROBLEMS

Perhaps the biggest error made in utilizing cyclic voltammetry is believing that this technique is the ideal method for determining the electrochemical properties of a redox system. Cyclic voltammetry is a powerful technique when applied to carefully chosen systems. However, little useful data will be acquired if the system of interest has excessive charge-transfer activation barriers or is heavily coupled to a complex chemical process. It is important to recall that

for an irreversible system, cyclic voltammetry provides no information about the redox potential—which is usually a central piece of information.

Aside from this "philosophical" problem, the experimenter needs also to be aware of several pragmatic pitfalls. Unlike in almost all other analytical characterization techniques, the "sample" is an active component of the instrument in an electrochemical reaction. Thus, the effective capacitances and resistances associated with the electrochemical cell actually modify the system response of the potentiostat. As such, artifactual signals or artifactual changes in signal are easily introduced into the study. Such signals can be mistakenly identified with chemical processes. It is therefore critical to know something about the resistive and capacitive response of the electrochemical cell under investigation, and it is important to minimize both of these elements. Usually the shape of the cyclic voltammogram or the behavior of the Nicholson and Shain diagnostics will give a clue that resistive components are affecting the observed signal. However, since the charge-transfer process in an irreversible system is identical to a physical resistance, artifactual resistances may not become obvious until they are large. Thus, cell design is of critical experimental importance. Excessively high impedance between the working and reference electrodes can lead to the total loss of peaks in the cyclic voltammogram. Such uncompensated resistance can arise from the use of high-resistance electrolytes; the placement of a junction separator between the working electrode and the reference electrode (i.e., in a multicompartment cell, the working and reference electrodes must be in the same compartment); junction potentials due to the presence of a different electrolyte in the reference half cell than employed in the cell; or a partial plugging of the junction separator. Thus, care must be taken in the selection of cell geometry and the choice of reference-electrode systems. The reader is directed to Gileadi (Gileadi et al., 1975; Gileadi, 1993) with regard to these details. In cases where a high uncompensated resistance cannot be avoided, various strategies can be implemented (Kissinger and Heineman, 1996). These include the use of a potentiostat with an iR compensation circuit and/or the addition of a Luggins capillary to the electrochemical cell. This latter device is a J-shaped capillary tube that holds the reference electrode and is placed as close as is possible to the working electrode (Gileadi et al., 1975; Gileadi, 1993).

In analyzing the response of a cyclic voltammetric cell, it is important to recognize that cyclic voltammetry is intrinsically a low-noise method. If the data is presenting a poor signal-to-noise ratio, then an instrumental error is involved. Two types of error need to be considered: (1) an actual failure in the potentiostat circuitry, and (2) an excessive (or variable) reference-electrode impedance. Noise is most often due to this second source. The impedance is induced by a partial blockage of the reference electrode junction. An alternate process that may introduce a noise-like condition is adsorption or desorption of reactants or products on the electrode surface. Surface adsorption can strongly distort cyclic voltammetric data. In certain cases, this process does not present itself as noise, but rather as an asymmetry in one of the cyclic voltammetric peaks. Variations in electrode response upon multiple scans of the potential window are also often due to adsorption processes. In cases where this occurs, the best solution is to find a solvent system or electrode material that does not exhibit this behavior. However, in some cases it is worth noting that adsorption can give rise to interesting materials properties. This phenomena has led to the development of the field of chemically modified electrodes and self-assembled surfaces, in which the surface properties of a material are tailored to provide specific chemical or physical responses to an external event. For example, highly selective and sensitive sensors can be developed by surface confining a molecular species to an electrode surface that interacts in a well-defined and unique manner with a target analyte. This type of device is illustrated by the sarcosine oxidase example provided earlier.

Finally, one needs to be aware that an artifactual response can be induced by instrumentation external to the electrochemical cell. When operating above ~500 mV/s, it is important to assure that electronics external to the cell are not limiting the potential slew rate. In particular, some potentiostat power supplies cannot be slewed sufficiently fast to achieve these rates. In addition, the recording device employed may not be able to keep up with high scan rates and may therefore produce artifactual results. This is particularly likely to be encountered if a mechanical x-y recorder is employed. However, limitations can also inadvertently be encountered with an analog-to-digital converter that interfaces the potentiostat output to a computer. For extremely high-speed experiments (>10 V/s), it is often necessary to observe the output on an oscilloscope.

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